Tuan-Binh Tran

Hanoi, Vietnam | binh.tt2@vinuni.edu.vn | Google Scholar | github.com/ttb06

Research Interest

My primary expertise lies in generative AI, specifically **Masked Diffusion Models** and **Flow Matching**. I am enthusiastic about exploring new research trajectories beyond my current focus on **recommender systems** and **time-series analysis**. Additionally, I am actively seeking industry research opportunities to tackle **practical, large-scale computational challenges**.

Education

VinUniversity Sep 2026 – Sep 2028
Master of Science in Computer Science (Expected)

University of Engineering and Technology, Vietnam National University, Hanoi Oct 2022 – Jun 2026
Bachelor of Science in Information Technology

- **Valedictorian (Joint rank 1st/192)** in Information Technology; GPA: 3.93/4.00.
- **Thesis:** Leveraging Large Language Models to Enrich Contextual Information in Time Series Analysis.

Publications

ScenarioDiff: A Scenario-level Guidance Framework for Multimodal Time Series Forecasting
Tuan-Binh Tran, Dat Nguyen Cong, Duc-Trong Le, Thanh Trung Huynh, Tung Kieu
2026 IEEE International Conference on Data Mining (ICDM'26), CORE A.* **[Preprint]**

EPIC: Explicit Posterior Item Conditioning for Semantic ID Diffusion Recommendation
Tuan-Binh Tran, Thanh Tam Nguyen, Quoc Viet Hung Nguyen, Duy Dung Le, Tung Kieu, Thanh Trung Huynh
Under review at the 20th ACM International Conference on Web Search and Data Mining (WSDM'27), CORE A. **[Preprint]**

Research Experience

Research Assistant, Center for AI Research, VinUniversity Feb 2026 – Present

- Advised by Dr. Thanh Trung Huynh, develop **discrete-diffusion recommender systems** with Semantic IDs, focusing on architecture, masking, long-identifier modeling, and ranking-oriented evaluation.

AI Residency Intern, FPT Software AI Center Jul 2025 – Feb 2026

- Advised by Dr. Tung Kieu, developed **multimodal time-series forecasting models** conditioned on textual context and analyzed when such context improves predictions.
- Built an **agentic predictive-maintenance pipeline** for explainable fault detection and classification.

Research Assistant, the Department of Computer Science, VNU–UET Mar 2024 – Jun 2025

- Conducted research and development on **continual learning algorithms** for vision-language models, focusing on architecture-based methods.

Awards and Honors

The 2023 ICPC Vietnam National Programming Contest Third Prize
Vietnam Mathematical Olympiad Third Prize (2022); Consolation Prize (2021)
VNU–UET Student Mathematics Olympiad 2024 Third Place
Inspiring Scholarship ($\times 5$ semesters)
2024 Posco Asia Fellowship
2023 MB Bank Scholarship

Academic Service

Subreviewer: CIKM'26

Techinical Skills

Programming/Tools: Python, C/C++, PyTorch, TensorFlow, Git, Linux, Shell, LaTeX.

Languages: Vietnamese (Native), English (IELTS 6.5), Simplified Chinese (Beginner).

References

Dr. Thanh Trung Huynh, Assistant Professor, College of Engineering and Computer Science, VinUniversity, Email: trung.ht@vinuni.edu.vn, Website: `https://thanhtrunghuynh93.github.io/`

Dr. Tung Kieu, Associate Professor, Department of Computer Science, Aalborg University, Email: tungkvt@cs.aau.dk, Website: `https://tungk.github.io/`.

Dr. Duc-Trong Le, Lecturer (Assistant Professor), Department of Computer Science, VNU–UET, Email: trongld@vnu.edu.vn. Website: `https://sites.google.com/view/trongld/`